

Project Title: Smart Hospital Management System

Project Overview:

The Smart Hospital Management System is a web-based solution designed to streamline patient care, appointment scheduling, medical records, and administrative workflows in healthcare institutions. The platform will be used by admin staff, doctors, patients, lab technicians, and pharmacists.

Project Metadata:

- **Project Type:** Web Application
 - **Time Constraint:** 3 months
 - **Project Scale:** Medium
 - **Project Budget:** ₹5,000
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Use Case Descriptions:

Use Case 1: Register New Patient

- **Actor:** Admin Staff
- **Description:** Allows the admin to register a new patient in the system by entering personal details, contact information, and insurance coverage.

Use Case 2: Book Appointment

- **Actor:** Patient
- **Description:** Patient can book appointments with doctors by selecting department, date, time slot, and preferred doctor (if available).

Use Case 3: Manage Appointments

- **Actor:** Doctor / Admin
- **Description:** View, accept, reschedule, or cancel appointments. Admin can override schedules in case of emergencies.

Use Case 4: Electronic Health Records (EHR) Management

- **Actor:** Doctor

- **Description:** Doctors can create and update patient health records including prescriptions, diagnosis notes, and lab reports.

Use Case 5: Prescription Upload

- **Actor:** Doctor
- **Description:** After consultation, doctors can generate and upload e-prescriptions to the patient's profile.

Use Case 6: Lab Test Booking

- **Actor:** Patient
- **Description:** Patients can request or schedule lab tests recommended by their doctor. System will automatically assign lab technician and generate test ticket.

Use Case 7: Lab Report Upload

- **Actor:** Lab Technician
- **Description:** After test completion, lab technician uploads the lab report which is then linked to the respective patient and accessible by the doctor.

Use Case 8: Pharmacy Order Management

- **Actor:** Pharmacist
- **Description:** Pharmacist can view prescribed medications, manage in-stock inventory, and mark medication as dispensed.

Use Case 9: Payment and Billing

- **Actor:** Patient / Admin
- **Description:** Patients can view and pay bills online for consultations, tests, or medication. Admin generates invoices and tracks payments.

Use Case 10: Feedback and Rating

- **Actor:** Patient
- **Description:** Patients can submit feedback and rate doctors, lab services, or overall hospital experience after each visit.

Additional Features (For Realism):

- **User Roles:** Multi-role login system (Admin, Doctor, Patient, Technician, Pharmacist)
- **Notifications:** SMS/email notifications for appointment reminders and test results
- **Data Security:** Role-based access, encryption for health records
- **Reports Dashboard:** Admin dashboard with analytics like daily appointments, income reports, service ratings